Final Exam

● Graded

Student

Ruiyang Hu

Total Points

42 / 45 pts

Question 1

Overall Grades **42** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 3 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Ruiyang Hu	rh403	胡瑞洋

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 胡瑞洋

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	4	5	3	5	5	42

Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

the system can be expressed as $\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow{2R_2-3R_3\to R_3} \left[\begin{array}{ccc|c} 4 & 0 & 9 & -1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow[R_3\to R_1]{R_1-4R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & 1 & -5 \end{array}\right] \xrightarrow[-R_2\to R_2]{R_1-2R_3\to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 2 & 3 & -1 \\ 0 & 0 & 1 & -5 \end{array}\right]$

$\xrightarrow{R_2-3R_3\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 2 & 0 & 14 \\ 0 & 0 & 1 & -5 \end{array}\right] \xrightarrow{\frac{1}{2}R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$

so we can get $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) the column space $\left\{\begin{bmatrix}2\\-5\\0\\1\end{bmatrix}, \begin{bmatrix}4\\0\\1\\3\end{bmatrix}, \begin{bmatrix}2\\5\\1\\2\end{bmatrix}\right\}$

as $\begin{bmatrix}2\\5\\1\\2\end{bmatrix} = \begin{bmatrix}4\\0\\1\\3\end{bmatrix} - \begin{bmatrix}2\\-5\\0\\1\end{bmatrix}$, (redundant) it can be simplified as $\left\{\begin{bmatrix}2\\-5\\0\\1\end{bmatrix}, \begin{bmatrix}4\\0\\1\\3\end{bmatrix}\right\}$

$a_1\begin{bmatrix}2\\-5\\0\\1\end{bmatrix} + a_2\begin{bmatrix}4\\0\\1\\3\end{bmatrix} = 0$ $\Rightarrow$ $\begin{cases} 2a_1+4a_2=0 \\ -5a_1=0 \\ a_2=0 \\ a_1+3a_2=0 \end{cases}$ only has trivial solution

the basis can be $\left\{\begin{bmatrix}2\\-5\\0\\1\end{bmatrix}, \begin{bmatrix}4\\0\\1\\3\end{bmatrix}\right\}$ OK

(2) row space $\left\{\begin{bmatrix}2\\4\end{bmatrix}^T, \begin{bmatrix}-5\\0\end{bmatrix}^T, \begin{bmatrix}0\\1\end{bmatrix}^T, \begin{bmatrix}1\\3\end{bmatrix}^T, \begin{bmatrix}2\\2\end{bmatrix}^T\right\}$

simplify to $\left\{\begin{bmatrix}2\\4\end{bmatrix}^T, \begin{bmatrix}1\\0\end{bmatrix}^T, \begin{bmatrix}0\\1\end{bmatrix}^T, \begin{bmatrix}1\\2\end{bmatrix}^T\right\}$

$\begin{bmatrix}1\\2\end{bmatrix}^T = \begin{bmatrix}1\\1\end{bmatrix}^T + \begin{bmatrix}0\\1\end{bmatrix}^T$ (redundant), simplify to $\left\{\begin{bmatrix}1\\2\end{bmatrix}^T, \begin{bmatrix}1\\0\end{bmatrix}^T, \begin{bmatrix}0\\1\end{bmatrix}^T\right\}$

$\begin{bmatrix}0\\1\end{bmatrix}^T = \frac{1}{2}\left(\begin{bmatrix}2\\1\end{bmatrix}^T - \begin{bmatrix}1\\0\end{bmatrix}^T\right)$ (redundant), simplify to $\left\{\begin{bmatrix}1\\2\end{bmatrix}^T, \begin{bmatrix}1\\0\end{bmatrix}^T\right\}$

$a_1\begin{bmatrix}1\\2\end{bmatrix}^T + a_2\begin{bmatrix}1\\0\end{bmatrix}^T = 0 \Rightarrow \begin{cases} a_1+a_2=0 \\ 2a_1=0 \\ a_1-a_2=0 \end{cases}$ only has trivial solution

the basis can be $\left\{\begin{bmatrix}1\\2\end{bmatrix}^T, \begin{bmatrix}1\\0\end{bmatrix}^T\right\}$

(3) having $\vec{u} = \begin{bmatrix}x\\y\\z\end{bmatrix}$ $A\vec{u}=0$ can be solved to $\begin{cases} x=z \\ y+z=0 \end{cases}$

so a basis of can be $\left\{\begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

suppose $A = [\vec{w_1}, \vec{w_2}, \vec{w_3}]$, A is the matrix which columns are $\vec{w_1}, \vec{w_2}, \vec{w_3}$

as $\vec{v} = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$, $T(\vec{v}) = x_1\vec{w_1} + x_2\vec{w_2} + x_3\vec{w_3}$

suppose $\vec{w_n} = \begin{bmatrix}a_n\\b_n\end{bmatrix}$, we have $\begin{bmatrix}a_2-a_3\\b_2-b_3\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$, $\begin{bmatrix}a_1-a_2\\b_1-b_2\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$, $\begin{bmatrix}-a_3\\-b_3\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$

solve and we can get $\vec{w_1} = \begin{bmatrix}-2\\0\end{bmatrix}$, $\vec{w_2} = \begin{bmatrix}-1\\-1\end{bmatrix}$, $\vec{w_3} = \begin{bmatrix}-2\\-2\end{bmatrix}$

so that $A = \begin{bmatrix}-2 & -1 & -2\\0 & -1 & -2\end{bmatrix}$

so $T\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}-2 & -1 & -2\\0 & -1 & -2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

we first need the eigenvalue λ

$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$ # $\det(A-\lambda I) = \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda)^2 = 0$

so that $\lambda_1 = 1$, multiplicity $= 2$, basis: $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$\lambda_2 = 2$, multiplicity $= 1$, basis: $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

so $J_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $J_2 = [2]$

$J = \begin{bmatrix} J_1 & 0 \\ 0 & J_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$A = PJP^{-1} = \begin{bmatrix} 0 & -1 & 2 \\ 0 & 0 & 1 \\ 2 & 1 & -6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -1 & 3 \\ -1 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

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Problem 5. [5] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \qquad A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$

$\det(A - \lambda I) = (3-\lambda)^2(4-\lambda) - (3-\lambda) - (3-\lambda)$

$= -(\lambda - 3)(\lambda - 2)(\lambda - 5)$

$\lambda_1 = 2. \quad \vec{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$\lambda_2 = 3 \quad \vec{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\lambda_3 = 5 \quad \vec{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

$\Rightarrow A = PDP^{-1}$

$P = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$ — 1

$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

so that $f(x,y,z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}$

$= f(u,v,w) = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} u & v & w \end{bmatrix}$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases}$$

transform to $\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_1+R_3 \to R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & 1 & 0 & 5 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_2+R_3 \to R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 3 & 0 & -3 \\ -2 & 2 & 0 & -8 \end{array}\right]$

$-R_1 \to R_3$, $\frac{1}{3}R_2 \to R_2$, $-\frac{1}{2}R_3 \to R_1$: $\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & -2 & 1 & 3 \end{array}\right] \xrightarrow[R_3+2R_2 \to R_3]{R_1+R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$

so that $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

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Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$V \subseteq \mathbb{R}^4$

use the Grand-Schimide Method.

the basis is $\{\vec{u_1}, \cdots \vec{u_n}\}$

$\vec{u_1} = \vec{v_1} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$\vec{u_2} = \vec{v_2} - \frac{\vec{u_1}\cdot\vec{v_2}}{\|\vec{u_1}\|^2}\vec{u_1}$

$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{(\sqrt{2})^2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$= \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

the orthonormal basis is $span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}\right\}$

only orthogonal

not orthonormal -2

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Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

we have $A^T = -A$. so A is anti-symmetric
since $A^T = -A$
so that $\det(A^T) = \det(-A)$
we have $\det(A^T) = \det(A)$
so that $\det(A) = \det(-A)$

~~as we multiply every row of A by k, is~~

we also have $\det(kA) = k^n \det(A)$
so $\det(A) = (-1)^n \det(A)$
since n is odd, $(-1)^n = -1$
so $\det(A) = -\det(A)$
so that $\det(A) = 0$

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

since $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

then $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$

so that $\vec{u}^2 + 2\vec{u}\cdot\vec{v} + \vec{v}^2 = \vec{u}^2 - 2\vec{u}\cdot\vec{v} + \vec{v}^2$

so that $\vec{u}\cdot\vec{v} = 0$

(i) If $\vec{u}$ and $\vec{v}$ are non-zero vectors,

$\vec{u}\cdot\vec{v} = \|\vec{u}\|\cdot\|\vec{v}\|\cos\theta = 0$. so $\cos\theta = 0$, so $\theta = \frac{\pi}{2}$, $\vec{u}$ and $\vec{v}$ are perpendicular

(ii) If $\vec{u}$ or $\vec{v}$ is zero vector,

by definition. zero vector is perpendicular to any other vector

In all. $\vec{u}$ and $\vec{v}$ are ⊥ perpendicular to each other